TEXORA · LECTURE SHEET

Comfort Parameter Test

The instrumental battery for comfort: sweating guarded hotplate (Rct and Ret), MVTR, air permeability, thermal absorptivity, moisture management and tactile evaluation — what each instrument measures, its governing standard, and worked results with their comfort interpretation.

AM-4103 · CLOTHING COMFORT & SPECIAL APPAREL PRODUCTS · TOPIC 2 · LEVEL 4 TERM 1

1. From perception to parameters

Topic 1 ended with perception, which cannot be specified in a purchase order. This topic converts each comfort type into **measurable fabric parameters**, each with an apparatus, a standard method and a unit. The mapping is the course's working table:

COMFORT ASPECT	PARAMETER	INSTRUMENT / STANDARD	UNIT
Dry insulation	Thermal resistance R_{ct}	Sweating guarded hotplate, ISO 11092	m²·K/W (tog, clo)
Moisture vapour escape	Water-vapour resistance R_{et}	Sweating guarded hotplate, ISO 11092	m²·Pa/W
Vapour throughput (alternative)	MVTR	Upright/inverted cup, ASTM E96; ISO 15496	g/m²/24 h
Ventilation	Air permeability	ISO 9237 (100 Pa, 20 cm²)	mm/s
Warm/cool touch	Thermal absorptivity b	Alambeta (and KES Thermo Labo)	$W \cdot s^{½}/m^2 \cdot K$
Liquid moisture handling	Wetting time, spreading speed, one-way transport index	Moisture Management Tester (AATCC 195)	s, mm/s, index
Handle / tactile	Bending, shear, compression, friction, roughness	KES-FB system; FAST; Handle-O-Meter	various
Drape and formability	Bending length, drape coefficient, formability	Cusick drape meter; FAST	cm, %, mm²

2. The skin model: sweating guarded hotplate (ISO 11092)

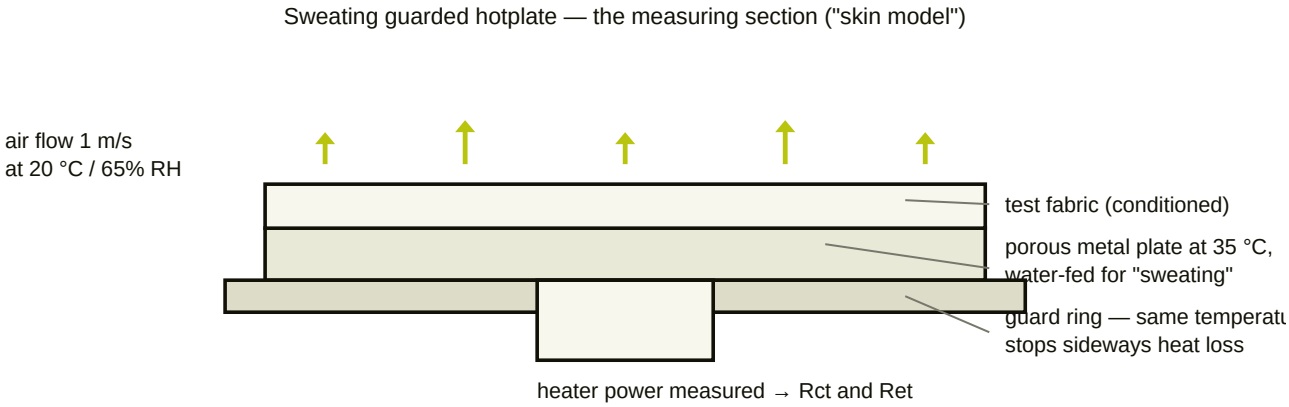


Figure 1 — The ISO 11092 skin model. Power needed to hold the plate at skin temperature, dry and sweating, gives R_{ct} and R_{et} after subtracting the bare-plate (air layer) values.

The plate is held at **35 °C** (skin temperature) in air at 20 °C, 65% RH, flowing at 1 m/s. Dry, the electrical power holding temperature gives total resistance; subtracting the bare-plate reading isolates the fabric's R_{ct}. With the plate fed water through its porous surface ("sweating") and covered by a vapour-permeable film, the power gives the evaporative resistance R_{et}. The Hohenstein comfort classification of R_{et} is the trade's shorthand:

R_{ET} (M²·PA/W)	< 6	6–13	13–20	20–30	> 30
Breathability verdict	Very good	Good	Satisfactory	Unsatisfactory	Very poor

WORKED EXAMPLES — VERIFY EVERY NUMBER

(a) MVTR, upright cup (ASTM E96). Cup opening diameter 8.0 cm → area = π × 0.04² = 0.00503 m². Mass loss over 24 h = 12.5 g. MVTR = 12.5 ÷ 0.00503 = **2,487 g/m²/24 h** — a respectably breathable shell fabric figure.

(b) Air permeability (ISO 9237). Flow 12.5 L/min through 20 cm² at 100 Pa. Flow = 12.5 × 1,000 ÷ 60 = 208.3 cm³/s. Permeability = 208.3 ÷ 20 = 10.4 cm/s = **104 mm/s**.

(c) Ret interpretation. Two shell fabrics: A, R_{et} = 8.5 m²·Pa/W → class "good"; B, R_{et} = 24 → "unsatisfactory". For the same wearer at the same work rate, B traps roughly three times the evaporative resistance — expect earlier sweat accumulation and wet cling, whatever B's waterproof rating says.

3. The remaining instruments in brief

Alambeta presses a heated head onto the fabric and records heat flow with time: it reports thermal conductivity, resistance and **thermal absorptivity b** — the warm/cool first-touch parameter of Topic 1 (typical b: wool ≈ 60–90 feels warm, cotton ≈ 180–220 mid, smooth synthetics 250+ feel cool; values are instrument-relative and compared within a study). **Moisture Management Tester** doses saline onto the fabric between sensor rings and reports wetting time of each face, absorption rate, spreading speed and the **one-way transport index** — the number that separates a true wicking knit (liquid pumped outward) from an absorbent one (liquid held at the skin). **KES-FB** measures the mechanical set — tensile, shear, bending, compression, surface friction (MIU) and roughness (SMD) — from which handle values and, via the Kawabata equations, comfort-relevant judgements like prickle risk and cling are built; **FAST** covers the same ground more cheaply for tailoring (formability, bending, extensibility). Topic 4 assembles these into product standards.

4. Reading a comfort test report critically

- **Conditioning:** results are valid only at the stated atmosphere (usually 20 °C, 65% RH) after conditioning; a Ret measured dry-conditioned differs from a sweaty in-use value.
- **Single-layer trap:** fabrics are tested flat and single; garments add air gaps, seams, closures and layering that change both R_{ct} and R_{et} — manikin tests (Topic 4) exist for that reason.
- **Direction and face:** wicking and permeability can differ face-to-back and warp-to-weft; a report without orientation is incomplete.
- **After-wash values:** comfort finishes (softeners, DWR) change Ret, absorptivity and wicking; specifications should state the wash count at test.

Glossary

TERM	MEANING
R_{et}	Water-vapour (evaporative) resistance, ISO 11092; lower is more breathable.
MVTR	Moisture vapour transmission rate through a specimen under cup-test conditions, g/m²/24 h.
Thermal absorptivity (b)	Surface heat-flow property deciding warm vs cool first touch.
One-way transport index	MMT measure of liquid moved from the skin face to the outer face; high is desirable next to skin.
Guarded hotplate	Hotplate whose guard ring is held at plate temperature so all measured power crosses the specimen.

Key points and summary

- Every comfort type maps to parameters with named instruments and standards — ISO 11092 (Rct/Ret) is the anchor test.
- Ret classes: < 6 very good to > 30 very poor breathability.
- MVTR and air permeability are simple area/flow calculations — always verify the arithmetic on a report.
- Instrument values are proxies: orientation, conditioning, layering and wash state decide whether they predict wear.

Practice

Answer in your own words; brief model answers follow.

1. A cup of opening diameter 7.0 cm loses 9.8 g in 24 h. Calculate the MVTR.
2. Air flow of 9.0 L/min passes a 20 cm² specimen at 100 Pa. Calculate the air permeability in mm/s.
3. A fabric's Ret is 17.5 m²·Pa/W. Give its Hohenstein class and predict the wearer's moisture experience in moderate work.
4. Why does the guarded hotplate subtract a bare-plate reading?
5. A sportswear brief says "high MVTR, low one-way transport index". Why is that combination wrong for a next-to-skin shirt?

Answer key

1. Area = $\pi \times 0.035^2 = 0.00385$ m². MVTR = $9.8 \div 0.00385 = 2{,}546$ g/m²/24 h.
2. Flow = $9.0 \times 1{,}000 \div 60 = 150$ cm³/s. Permeability = $150 \div 20 = 7.5$ cm/s = 75 mm/s.
3. Class "satisfactory" (13–20). Vapour escapes, but at moderate work sweat will begin to accumulate in the microclimate during sustained effort; acceptable for low-to-moderate activity, marginal for high output.
4. The measured power crosses fabric + boundary air layer in series. The bare-plate run measures the air layer (and apparatus) alone; subtracting it leaves the fabric's own resistance.
5. MVTR moves vapour, but liquid sweat at the skin needs one-way liquid transport away from the body. Low one-way index means liquid stays at the skin face: wet cling and chilling, despite good vapour figures.